Orbital Motion and Energy Drift: Exercise Guide for 2C

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Mathematical Methods for Physicists II

Coursework Notes and Computational Guides

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This is a collection of notes from my Mathematical Methods for Physicists II coursework on orbital motion, coupled differential equations, and conservation laws. It accompanies notebook 2C and explains the gravitational and orbital concepts before walking through the project. Numerical methods already developed in the shared theory guide for 2A and 2B are referenced rather than repeated. These notes support learning and independent implementation; the orbit is a model, not an astronomical ephemeris.

1 Prerequisites and the question of this project

The shared guide *Oscillators, Pendulums, and Numerical Dynamics* introduces initial-value problems, coupled-state integration, Euler, midpoint RK2, RK4, velocity Verlet, and numerical refinement. Those derivations are not repeated here. Consult its sections on accuracy, stability, and conservation diagnostics when needed.

The new problem is planar orbital motion under Newtonian gravity. Four methods solve the same equations from the same initial state. The question is whether changes in the computed orbit reflect physical dynamics or numerical error. To answer it we need gravitational acceleration, orbital energy, angular momentum, and the relationship between conserved quantities and orbital shape.

The guide first develops those ideas, then follows the notebook's model, trajectory plots, conservation plots, and refinement table. No probability or random forcing is needed. Despite the nonlinear force, this isolated Kepler problem is integrable; numerical drift here is not a demonstration of physical chaos.

2 Newtonian gravity and the test-particle model

2.1 From force to vector acceleration

A fixed star of mass M exerts a force of magnitude GMm/r^2 on a planet of mass m. The force points toward the star. With the star at the origin, let

$$\mathbf{r} = (x, y), \quad r = \sqrt{x^2 + y^2}, \quad \hat{\mathbf{r}} = \mathbf{r}/r.$$

Then

$$\mathbf{F} = -\frac{GMm}{r^2}\hat{\mathbf{r}}, \qquad \mathbf{a} = -\frac{\mu \mathbf{r}}{r^3}, \qquad \mu = GM.$$

The denominator has a third power because the numerator is a vector of length r: the acceleration magnitude still varies as r^{-2}. In Cartesian components,

$$a_x = -\frac{\mu x}{(x^2 + y^2)^{3/2}}, \qquad a_y = -\frac{\mu y}{(x^2 + y^2)^{3/2}}.$$

Both components depend on the same distance, so the two coordinate equations are coupled.

2.2 What is being idealized

The planet is a test particle: its mass cancels from acceleration and its effect on the star is neglected. The star stays at the origin. There are no other planets, drag, thrust, collisions, or relativistic corrections. Motion stays in a plane because a central force preserves the orbital plane.

For two finite masses moving about their common centre of mass, the relative coordinate instead obeys the two-body equation with $\mu = G(M + m)$. That is not the model used here. Likewise the momentum of the planet alone is not conserved, because gravity changes its direction. Specific orbital energy and angular momentum about the fixed star are the relevant invariants.

2.3 The singularity

The point-mass force is undefined at $r = 0$ and grows sharply at small r. Do not replace it silently with a softened force; softening changes the physical model. The chosen initial conditions have nonzero angular momentum and a periapsis well away from zero. A collision or very close-passage project would require additional numerical treatment.

3 Units and the gravitational parameter

The notebook measures position in astronomical units (AU) and time in years. It chooses

$$\mu = 4\pi^2 \ \mathrm{AU}^3\,\mathrm{yr}^{-2}.$$

Thus velocity has units AU/yr and acceleration AU/yr^2. The number $4\pi^2$ is a convenient solar-like normalization, not a measured ephemeris parameter.

For a circular orbit, centripetal acceleration gives

$$\frac{v_c^2}{r} = \frac{\mu}{r^2}, \qquad v_c = \sqrt{\mu/r}.$$

The period is circumference divided by speed:

$$P_c = 2\pi\sqrt{r^3/\mu}.$$

At $r = 1$ AU, these choices give $v_c = 2\pi$ AU/yr and $P_c = 1$ yr. This explains the numerical initial speed in the code. Do not insert SI values of G into an AU/year calculation without converting units.

4 Specific energy: bound motion and numerical gain

4.1 Derive the conserved quantity

The gravitational potential per unit planet mass is $u = -\mu/r$, with zero at infinite distance. The specific mechanical energy is

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r}.$$

Its units are AU^2/yr^2, and total energy would be $m\varepsilon$ in compatible units. The mass has not been set to one physically; we have divided energy by it.

Differentiate along a trajectory:

$$\frac{d\varepsilon}{dt} = \mathbf{v}\cdot\mathbf{a} + \frac{\mu\,\mathbf{r}\cdot\mathbf{v}}{r^3} = -\frac{\mu\,\mathbf{r}\cdot\mathbf{v}}{r^3} + \frac{\mu\,\mathbf{r}\cdot\mathbf{v}}{r^3} = 0.$$

Kinetic and potential energies individually vary as the planet moves, but their sum remains constant. A faster planet near the star has more kinetic energy and a more negative potential energy, not an extra source of mechanical energy.

4.2 Interpret the sign

For a noncollision Kepler orbit, negative ε corresponds to a bound ellipse, zero to a parabolic escape threshold, and positive energy to an unbound hyperbola. Escape speed at distance r follows from $\varepsilon = 0$:

$$v_{\text{esc}} = \sqrt{2\mu/r} = \sqrt{2}\,v_c.$$

A numerical energy increase means a less negative energy while the orbit remains bound; it does not automatically imply escape. In this notebook Euler's energy rises substantially but is still negative at the final time.

5 Specific angular momentum and central forces

5.1 Derive the second invariant

Specific angular momentum is $\boldsymbol{\ell} = \mathbf{r}\times\mathbf{v}$. In the planar calculation only the perpendicular component is nonzero:

$$\ell = xv_y - yv_x.$$

Its units are AU^2/yr. Positive ℓ indicates counterclockwise motion in the chosen coordinates. This is angular momentum per unit planet mass, not angular velocity.

Differentiate:

$$\dot{\boldsymbol{\ell}} = \mathbf{v}\times\mathbf{v} + \mathbf{r}\times\mathbf{a} = 0.$$

The cross product vanishes because the acceleration is parallel to the radius. Equivalently a central force produces no torque about the origin.

5.2 Geometric meaning: equal areas

In polar coordinates $\ell = r^2\dot{\varphi}$, where φ is the orbital angle. The swept area rate is $\dot{A} = \ell/2$. Constant angular momentum therefore gives Kepler's equal-areas law. The planet moves through angle more rapidly when it is closer to the star.

Energy and angular momentum are independent diagnostics. One constrains the orbital energy scale, the other the transverse motion. Their conservation is necessary for the correct orbit, but does not guarantee that the numerical position is correct at a given time.

6 Ellipse geometry and the chosen initial state

6.1 Orbital elements from the invariants

For a bound Kepler ellipse,

$$a = -\frac{\mu}{2\varepsilon}, \qquad e = \sqrt{1 + \frac{2\varepsilon\ell^2}{\mu^2}}, \qquad p = \frac{\ell^2}{\mu} = a(1 - e^2).$$

Here a is the semimajor axis, e the eccentricity, and p the semilatus rectum. The periapsis and apoapsis distances are

$$r_p = a(1 - e), \qquad r_a = a(1 + e).$$

Periapsis is the nearest point to the star; apoapsis is the farthest. For a solar-centred orbit these are also called perihelion and aphelion. The orbital period is

$$P = 2\pi\sqrt{a^3/\mu}.$$

These relations can be derived by reducing the central-force equation to a radial equation and solving the inverse-square orbit; they provide checks without integrating numerically.

6.2 Calculate this orbit before coding

The initial conditions are

$$\mathbf{r}_0 = (1, 0)\ \text{AU}, \qquad \mathbf{v}_0 = (0, 0.9 \times 2\pi)\ \text{AU/yr}.$$

The initial radial velocity is zero, and the tangential speed is below circular speed. The starting point is therefore apoapsis, not periapsis. Write $f = 0.9$ for the circular-speed fraction. Then

$$\varepsilon_0 = (f^2/2 - 1)\mu = -0.595\mu \approx -23.48966, \qquad \ell_0 = 2\pi f \approx 5.65487.$$

The implied elements are

$$a = \frac{1}{2 - f^2} \approx 0.840336\ \text{AU}, \quad e = 1 - f^2 = 0.19, \quad r_a = 1\ \text{AU},$$

$$r_p \approx 0.680672\ \text{AU}, \quad p = f^2 = 0.81\ \text{AU}, \quad P \approx 0.770335\ \text{yr}.$$

Thus the exact orbit stays within 1 AU of the star. A computed path reaching 10 AU is a major numerical departure. Twenty years cover about 26 physical orbital periods, making accumulated error visible.

6.3 A reference shape and orientation

Measuring φ from the positive x axis, the ellipse beginning at apoapsis obeys

$$r(\varphi) = \frac{p}{1 - e\cos\varphi}.$$

At $\varphi = 0$ this gives 1 AU, and at $\varphi = \pi$ it gives the periapsis on the negative x axis. The star is at a focus, not the geometric centre. This formula is a spatial reference; it does not supply the position at a specified time without also solving the time dependence.

7 Radial motion and why close passages are harder

Resolve velocity into radial and tangential parts. Since $v_{\text{tangential}} = \ell/r$,

$$\varepsilon = \tfrac{1}{2}\dot{r}^2 + u_{\text{eff}}(r), \qquad u_{\text{eff}}(r) = \frac{\ell^2}{2r^2} - \frac{\mu}{r}.$$

The positive centrifugal term produces a barrier at small r for nonzero angular momentum. Turning points satisfy $\dot{r} = 0$ and $\varepsilon = u_{\text{eff}}$, giving periapsis and apoapsis.

At periapsis the angular rate $\dot{\varphi} = \ell/r^2$ is largest. Here it is about 12.2 rad/yr, so the main step $h = 0.025$ yr advances roughly 0.305 rad in a periapsis step. A step that looks small relative to the whole period may still be coarse where the force changes fastest. More eccentric orbits need finer resolution near closest approach.

The familiar harmonic stability condition from 2A is not a universal orbit-accuracy criterion. Resolve local motion and test refinement rather than transferring that threshold directly to an inverse-square force.

8 What conservation does and does not validate

8.1 Energy drift changes the orbit

For negative energy, $a = -\mu/(2\varepsilon)$. Making ε less negative increases the corresponding Kepler semimajor axis and period. Changing ℓ also changes the inferred eccentricity. Consequently a widening numerical orbit can be explained by errors in its invariants.

One may compute these elements from each instantaneous numerical state as *osculating elements*: the Kepler orbit that would match that position and velocity. They are not evidence that a drifting numerical trajectory is an exact ellipse with a changing physical force. They diagnose the departure from the fixed conservative model.

8.2 Angular momentum in Euler and Verlet

Forward Euler uses $\mathbf{r}' = \mathbf{r} + h\mathbf{v}$ and $\mathbf{v}' = \mathbf{v} + h\mathbf{a}$. For central gravity,

$$\ell' = \ell + h^2(\mathbf{v} \times \mathbf{a})_z = \left(1 + \frac{\mu h^2}{r^3}\right)\ell.$$

Thus it artificially increases positive angular momentum at every step for this orbit. The torque-free continuous model does not have this gain.

Velocity Verlet can be viewed as a half velocity kick, a position drift, and a second half kick. A central-force kick does not change $\mathbf{r} \times \mathbf{v}$; a drift does not change it either because $\mathbf{v} \times \mathbf{v} = 0$. Consequently this implementation preserves angular momentum up to floating-point roundoff. Its energy is nevertheless approximate and can oscillate.

8.3 Shape, phase, and numerical precession

A method can have nearly constant energy and angular momentum yet accumulate orbital phase error. A numerical orbit can also show a slowly changing periapsis direction, called numerical precession here. The exact Newtonian two-body ellipse has a fixed orientation. This numerical effect is not relativistic precession or evidence for another planet.

An additional Kepler invariant is the eccentricity vector,

$$\mathbf{e} = \frac{(v^2 - \mu/r)\mathbf{r} - (\mathbf{r} \cdot \mathbf{v})\mathbf{v}}{\mu}.$$

It points toward periapsis and has magnitude e; initially it is $(-0.19, 0)$. This offers a future orientation diagnostic, but it is not calculated in the present notebook. Full trajectory or orbital-phase checks would strengthen validation beyond the two existing invariants.

9 Walkthrough 1: assemble the model and solver

9.1 Design the four-component state

Use the ordered array $\mathbf{y} = (x, y, v_x, v_y)$. Its derivative is

$$\dot{\mathbf{y}} = (v_x, v_y, a_x, a_y).$$

In the code, `state[:2]` extracts position and `state[2:]` extracts velocity. The slope function concatenates velocity with acceleration. The norm in the acceleration function is the norm of the position vector only; using the four-component state would incorrectly mix distance and speed.

Reuse the methods explained in the shared theory guide. Euler evaluates one old-state slope. RK2 predicts the entire midpoint state. RK4 evaluates four coupled states. Verlet predicts position with old acceleration, recomputes gravity at the new position, and averages the accelerations for velocity. All use exactly the same gravity function.

9.2 Create a history

The main experiment uses $h = 0.025$ yr and duration 20 yr, hence 800 steps and 801 rows. Create a time array $t_i = ih$ and a zero-filled array of shape $(801, 4)$. Put the initial state in row zero. For every iteration, calculate the next row from the current row.

The duration divides exactly by the step for these choices. In a reusable solver check this explicitly: rounding T/h for an arbitrary duration can produce a different endpoint. Every method must start from the same initial state, not from the final state of the previous method.

9.3 Before making plots

Check the initial acceleration $\mathbf{a}_0 = (-\mu, 0)$, the initial invariant values, the number of rows, and the endpoint time. All saved values should be finite. A circular-orbit initial speed of 2π could supply a simple additional reference in a separate experiment, but changing to it would change the present scientific scenario.

10 Walkthrough 2: compare orbital shapes

For each method plot the saved x values against saved y values, mark the fixed star, and mark the starting point. Equal aspect ratio is essential: one unit on the horizontal axis must equal one unit on the vertical axis. Otherwise a circle can look elliptical merely because of plotting geometry.

The revised main panels use a common extent calculated from the largest absolute coordinate across *all* trajectories, with a small margin. This prevents Euler's large outward excursion from being hidden by the original limits of ± 3 AU. A close-up inset with limits ± 1.8 AU makes the smaller orbits readable. Parts outside an inset are deliberately outside its close-up, while the full path remains visible in the main panel.

For the given experiment Euler's maximum radius is about 10.39 AU. RK2 also expands noticeably, reaching about 1.66 AU. RK4 and Verlet remain near the expected spatial scale. These are outputs for this step and duration, not statements that every run of these methods has the same behaviour.

A curve that leaves its initial ellipse is a warning. Diagnose it with conservation histories rather than calling it a physical instability, escape, or gravitational perturbation. A position plot does not compare orbital phase at equal times; it overlays the path traced throughout the run.

11 Walkthrough 3: calculate and plot invariants

11.1 Evaluate every row at once

Extract the position matrix and velocity matrix from the saved array. For each row,

$$v_i^2 = v_{xi}^2 + v_{yi}^2, \quad r_i = \sqrt{x_i^2 + y_i^2}, \quad \varepsilon_i = v_i^2/2 - \mu/r_i, \quad \ell_i = x_i v_{yi} - y_i v_{xi}.$$

The code's sum with `axis=1` sums the two components within each row. The norm also uses `axis=1`, producing one distance per time. Summing over the whole array would erase the time history.

11.2 Why the normalization uses absolute initial values

Plot dimensionless changes

$$\delta_\varepsilon(t) = \frac{\varepsilon(t) - \varepsilon_0}{|\varepsilon_0|}, \qquad \delta_\ell(t) = \frac{\ell(t) - \ell_0}{|\ell_0|}.$$

For a bound orbit $\varepsilon_0 < 0$. Dividing the signed change by $|\varepsilon_0|$ makes positive plotted change mean energy gain. Dividing instead by ε_0 would reverse that sign. Absolute deviations have the same magnitude with either denominator, which explains the code's equivalent expression for peak error.

The notebook's variables `E` and `L` represent specific energy and specific angular momentum, not total energy and rod length. Both initial values are nonzero in this experiment, so the normalized diagnostics are well defined. A zero-energy or zero-angular-momentum scenario would require another error scale.

11.3 Interpret the actual comparison

The twenty-year outputs give approximately:

Method	Maximum $\lvert\delta_\varepsilon\rvert$	Maximum $\lvert\delta_\ell\rvert$
Euler	8.68×10^{-1}	1.09
Midpoint RK2	2.80×10^{-1}	8.35×10^{-2}
RK4	4.42×10^{-3}	1.64×10^{-3}
Velocity Verlet	8.99×10^{-3}	about 2×10^{-15}

Euler and RK2 gain artificial energy here. RK4 shows a smaller net energy loss. Verlet has a bounded-looking energy oscillation on this tested interval and nearly roundoff-level angular momentum change. Euler's large deviations dominate the shared vertical scale, so the printed energy statistics are needed to compare smaller errors.

A higher-order method and a symplectic method control errors in different ways. At this resolution RK4's peak energy error is smaller than Verlet's, while Verlet preserves angular momentum much better. This illustrates why one number is not a complete ranking. Do not claim a bounded energy error for every step size or indefinitely from this finite run alone.

12 Walkthrough 4: refine the step at fixed duration

The refinement experiment runs for eight years, using $h = 0.04, 0.02, 0.01$ yr. These give 200, 400, and 800 steps. It is deliberately a separate, shorter comparison than the twenty-year plots; compare step-size rows at the common eight-year duration.

For each step and method:

1. Integrate from the original initial state to eight years.
2. Calculate the energy at all saved states.
3. Report the peak absolute relative deviation, $\max_i |\varepsilon_i - \varepsilon_0|/|\varepsilon_0|$.
4. Report the signed final change, $(\varepsilon_N - \varepsilon_0)/|\varepsilon_0|$.

The peak includes oscillatory error and secular drift. A tiny endpoint change can occur when an energy oscillation returns close to zero; it does not mean the entire history was equally accurate.

The quantities reported are errors in a conserved observable, not trajectory errors against an exact solution. They need not display the methods' formal global position-error orders directly. Euler's coarse, strongly drifting trajectory also leaves the physical regime intended for an asymptotic convergence comparison.

Read refinement as a consistency test: does reducing the step reduce the relevant departures, and does the interpretation of the plotted orbit remain credible? Accurate invariants still do not certify correct orbital phase.

13 Equal step size is not equal computational cost

The current comparison holds h fixed, so all methods save states at the same times and use the same number of steps. They do not perform the same number of gravity evaluations. In this implementation Euler uses one per step, midpoint RK2 two, RK4 four, and Verlet two.

For position-dependent forces Verlet could reuse the final acceleration of one step as the initial acceleration of the next, reducing repeated work. The current function does not cache it. An equal-cost comparison should count the force evaluations actually performed, including initialization, rather than assuming every method costs one evaluation.

Equal step comparisons answer which method behaves better at a chosen time resolution. Equal-cost comparisons answer which method achieves better accuracy for a chosen computation budget. They are different experiments, and the notebook clearly performs the first.

14 A route to writing your own version

Write the model and units first. Choose a four-component state and a single gravity function. Test the slope's component order, then implement the numerical steps using the shared theory guide. Construct histories with an initial row, and independently evaluate specific energy and angular momentum from those histories.

Make full-scale orbit plots with equal aspect ratio. Inspect conservation histories, endpoint changes, and maximum deviations. Refine over a fixed duration and retain the distinction between a physical change in kinetic energy and an artificial change in total specific energy. If you extend the problem to a more eccentric orbit, prioritize resolving periapsis rather than merely keeping the old number of steps.

No change to the gravitational equations or solver formulas was necessary in the reviewed notebook. The substantive correction was making the full drift visible in the orbital plots. Its remaining limitations—finite-duration conservation checks, unequal cost per step, and no explicit phase-error benchmark—are part of interpreting the project's scope, rather than faults in the implemented experiment.

15 Book references

1. John R. Taylor, *Classical Mechanics*, University Science Books. Read the discussions of energy, angular momentum, central forces, and the Kepler problem for the physical foundations. Publisher information: https://uscibooks.aip.org/books/classical-mechanics/.
2. Carl D. Murray and Stanley F. Dermott, *Solar System Dynamics*, Cambridge University Press. The two-body problem provides a more advanced route into orbital elements and celestial mechanics. Author-hosted book information: https://ssdbook.maths.qmul.ac.uk/index.html.
3. William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd edition, Cambridge University Press, Chapter 17. Use this with the shared numerical theory guide for integration and error control. Official book contents: https://numerical.recipes/book/.

The notes are an independent explanation of the coursework, not copied textbook passages. Start with Taylor for the orbital physics; Murray and Dermott is a deeper reference rather than a prerequisite for implementing this project.